

Boosted Multi-Layer Edge Combination for Scientific Paper Clustering

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Abstract

We propose a multi-layer edge combination pipeline for CPM Leiden clustering of scientific paper networks. The pipeline combines citation-based edges (direct citation, bibliographic coupling, co-citation) with embedding-based edges using a *boosted consensus* strategy: edges confirmed by multiple independent layers receive multiplicative weight amplification. Combined with per-node top- k filtering, 1/rank normalization, and automatic resolution (γ) selection, the pipeline produces balanced cluster distributions across fields with varying citation densities. Experiments on two OpenAlex fields (Chemical Engineering, 115K papers; Arts & Humanities, 754K papers) demonstrate that (1) boosted combination assigns boundary nodes more accurately than simple summation (29:26 in 55 LLM blind reviews), (2) adding embedding edges reduces cluster count by 60% while doubling average cluster size, and (3) automatic γ search is essential as optimal resolution shifts $24\text{--}32\times$ between 3-layer and 4-layer configurations.

1 Introduction

Clustering scientific papers into research communities requires constructing a similarity network from multiple heterogeneous signals. Citation-based measures—direct citation (DC), bibliographic coupling (BC), and co-citation (CC)—capture different aspects of scholarly relatedness but have uneven coverage: BC may cover only 49% of papers in humanities fields, while DC is universally available but sparse. Embedding-based similarity (e.g., SPECTER2 k -NN) provides universal coverage but may introduce noise from stylistic rather than topical similarity.

The central question is: *how should these heterogeneous layers be combined into a single network for community detection?*

We address three sub-problems:

1. **Normalization:** Edge weights across layers have incomparable scales.
2. **Combination:** How to merge normalized layers into one edge table.
3. **Resolution:** The CPM resolution parameter γ must adapt to the combined network’s density.

2 Method

2.1 Pipeline Overview

2.2 Top- k Filtering (Step 1)

Each layer independently retains only the k strongest neighbors per node (symmetric mode: edge kept if *either* endpoint has it in their top- k). This removes long-tail noise that otherwise dominates after normalization. With $k=30$, cross-cluster edges drop from 70% to 31% of the combined network.

Algorithm 1 Boosted Multi-Layer Combination

Require: Edge tables $\{E_\ell\}_{\ell=1}^L$ for L layers, target max%

```
1: for each layer  $\ell$  do
2:    $E'_\ell \leftarrow \text{TOPK}(E_\ell, k=30)$  {per-node top- $k$  filter}
3:    $E''_\ell \leftarrow \text{RANKNORM}(E'_\ell)$   $\{w_{ij} \leftarrow 1/\text{rank}(w_{ij})\}$ 
4: end for
5:  $E_{\text{combined}} \leftarrow \text{BOOSTEDSUM}(\{E''_\ell\})$  {Eq. ??}
6:  $E_{\text{gcc}} \leftarrow \text{GCC}(E_{\text{combined}})$  {giant connected component}
7:  $\gamma^* \leftarrow \text{AUTOGAMMA}(E_{\text{gcc}}, \text{max}\%)$  {binary search}
8:  $\mathcal{C} \leftarrow \text{LEIDEN}(E_{\text{gcc}}, \gamma^*)$ 
9:  $\mathcal{C}' \leftarrow \text{POSTPROCESS}(\mathcal{C})$  {cascade + Dijkstra}
10: return  $\mathcal{C}'$ 
```

2.3 1/rank Normalization (Step 2)

Within each filtered layer, edges are sorted by weight descending and assigned $w_{ij} \leftarrow 1/\text{rank}(w_{ij})$. This is scale-free: it preserves only the ordering, making heterogeneous layers (BC cosine similarity vs. DC fractional counts vs. Emb k -NN distance) directly comparable.

2.4 Boosted Consensus Sum (Step 3)

$$w_{ij}^{\text{boost}} = \left(\sum_{\ell=1}^L w_{ij}^{(\ell)} \right) \times n_{ij} \quad (1)$$

where $n_{ij} = |\{\ell : (i, j) \in E'_\ell\}|$ is the number of layers containing edge (i, j) after top- k filtering. Edges confirmed by multiple independent signals receive multiplicative amplification.

Rationale. In a 3-layer (BC+CC+DC) configuration:

- 73% of edges appear in only 1 layer ($n_{ij}=1$, no boost)
- 21% appear in 2 layers ($n_{ij}=2$, 2 $\times$ boost)
- 6% appear in all 3 layers ($n_{ij}=3$, 3 $\times$ boost)

The 6% of 3-layer edges form *cluster cores* in Leiden, while single-layer edges become relatively weak, preventing mega-cluster absorption.

2.5 Automatic Resolution Selection (Step 4)

Adding layers changes the effective edge density, shifting optimal γ non-linearly. We observe 24–32 $\times$ shifts between 3-layer and 4-layer configurations—unpredictable from edge count ratios alone (4.2 $\times$). We therefore use binary search on log-scale γ to find the resolution where the largest cluster is $< \text{max}\%$ of total nodes (default 3%).

3 Experimental Setup

3.1 Data

Two OpenAlex fields with contrasting citation densities:

3.2 Comparison Methods

Normalization: raw, 1/rank, quantile.

Combination: sum ($\sum w$), max ($\max w$), vote (count), boosted ($\sum w \times n$).

Table 1: Dataset characteristics (top-30 filtered)

Layer	Field 15 (Chem. Eng.)		Field 12 (Arts & Hum.)	
	Edges	Coverage	Edges	Coverage
BC cosine	1.56M	86%	–	49%
CC cosine	524K	46%	–	18%
DC fractional	808K	100%	–	100%
Emb k -NN	2.86M	100%	19.3M	100%
DC avg degree	16		5	
Emb avg degree	50		51	

3.3 Evaluation

1. **Structural**: cluster count, max cluster %, size distribution balance.
2. **LLM blind review** (GPT-4o-mini, 55 cases): three criteria on disagreement nodes between sum and boosted:
 - *Belonging*: “Which group does this paper naturally belong to?”
 - *Group cohesion*: independent quality score per cluster (1–5).
 - *Outlier count*: papers that don’t fit in the cluster.

4 Results

4.1 Normalization $\times$ Combination Grid

Table 2: Field 15, $\gamma=10^{-6}$, min_size=100, top-30, GCC. Cluster count and max cluster %.

Normalization	Combination	Clusters	Max %
raw	sum	189	12.2
raw	vote	149	15.5
1/rank	sum	282	2.4
1/rank	boosted	298	2.5
1/rank	vote	149	16.5
quantile	sum	156	15.8

1/rank with sum or boosted produces the most balanced distributions. Vote is invariant to normalization (binary). Raw sum creates dominant clusters.

4.2 Sum vs. Boosted: LLM Evaluation

Table 3: Triple evaluation on disagreement nodes ($\gamma=10^{-4}$, Field 15).

Criterion	Sum	Boosted	Better
Belonging (wins, $n=55$)	26	29	Boosted
Cohesion (avg, 1–5)	4.4	4.4	Tie
Outliers (avg per 8 neighbors)	1.3	1.3	Tie

Boosted assigns boundary nodes more accurately. Internal cluster quality is equivalent.

By cluster size bucket (30 cases):

- **Large (>500)**: Sum 7:3 — sum’s mega-clusters win by volume.
- **Medium (200–500)**: Boosted 6:4 — consensus correctly separates.
- **Small (<200)**: Tie 5:5.

4.3 3-Layer vs. 4-Layer (+Embedding)

Table 4: Auto-gamma results (target max < 3%).

Config	Auto γ	Clusters	Max %	Avg size
Field 15, 3L	5.62×10^{-5}	251	3.0	459
Field 15, 4L	1.33×10^{-3}	221	2.8	521
Field 12, 3L	1.78×10^{-6}	1,157	2.9	651
Field 12, 4L	5.62×10^{-5}	445	2.9	1,693

Adding embedding edges consistently improves partitioning:

- Field 15: 12% fewer clusters, 13% larger avg size.
- Field 12: **62% fewer clusters, 160% larger avg size** — embedding fills citation gaps in humanities.

The γ ratio between 3L and 4L is $24\times$ (Field 15) to $32\times$ (Field 12), confirming that automatic search is essential.

4.4 Boosted Mechanism Analysis

The boosted/sum weight ratio is exactly $n_{ij} \in \{1, 2, 3, 4\}$:

Table 5: Edge distribution by layer count (Field 15, 3-layer).

n_{ij}	Fraction	Boost	Role
1	73%	$1\times$	Boundary/noise
2	21%	$2\times$	Supporting
3	6.5%	$3\times$	Cluster cores

The 6.5% of 3-layer edges, boosted to $3\times$ weight, dominate Leiden’s objective function and form cluster nuclei. Single-layer edges become relatively weak, preventing the mega-cluster absorption observed with simple summation.

5 Discussion

Why boosted works. CPM Leiden optimizes $H = \sum_c [e_c - \gamma \binom{n_c}{2}]$, where e_c is internal edge weight. Boosted consensus amplifies edges that multiple independent measures agree on, concentrating weight in true communities. This creates a natural hierarchy: multi-layer cores attract single-layer periphery.

Embedding layer integration. Embedding k -NN edges have universal coverage (avg degree ≈ 50) unlike citation layers (DC avg degree = 5 in humanities). Naïvely adding Emb to boosted increases n_{ij} globally, collapsing the differentiation. The solution is **not** to exclude Emb but to **automatically adjust** γ : auto-gamma search absorbs the density shift, letting Emb fill citation gaps while boosted consensus preserves cluster structure.

Field dependence. Citation-poor fields (Arts & Humanities: 3-layer overlap 0.1%) benefit most from Emb addition (160% avg size improvement) because citation layers alone produce excessive fragmentation. Citation-rich fields (Chemical Engineering: 3-layer overlap 2.2%) see moderate improvement (13%).

6 Conclusion

We recommend the following pipeline for multi-layer scientific paper clustering:

1. **Top-30** per-node filtering on each layer.
2. $1/\text{rank}$ normalization for scale-free comparison.
3. **Boosted consensus sum** ($\sum w \times n_{\text{layers}}$) for multi-layer agreement.
4. **GCC filtering** to remove isolated components.
5. **Auto-gamma** binary search targeting max cluster $< 3\%$.
6. **Leiden + postprocess** with cascading γ descent.

This pipeline adapts automatically to varying citation densities across fields and layer configurations, producing balanced cluster distributions without manual parameter tuning.